# Business Analytics

Final Project -- Student Instruction Guide

Course Project
All Phases
Individual / Group

## Project Overview

This project is designed to build your practical skills in business analytics through a full end-to-end analysis workflow. You will select a real-world dataset, define business questions, perform exploratory data analysis using Python, and communicate your findings through dashboards and visualizations.

The project emphasizes descriptive analytics -- understanding what the data says -- with a light predictive component to explore what the data suggests about future patterns.

**Core Focus Areas**

Descriptive Analytics | Exploratory Data Analysis (EDA) | Data Visualization | Dashboards | Basic Statistics | Light Predictive Analysis

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## Business Questions

Your project must address seven (7) business questions in total: five (5) main questions and two (2) ad hoc questions. Each main question must represent a distinct Business Analytics concept as listed below.

You are not required to use a specific dataset -- choose any real-world or publicly available dataset relevant to a business domain of your choosing (e.g., retail, healthcare, HR, finance, e-commerce). Make sure the dataset is rich enough to support all seven questions.

### Main Questions

**Main Question Q1 Descriptive Statistics**

Summarize the key variables in your dataset. Report measures of central tendency (mean, median, mode), spread (standard deviation, variance, range, IQR), and shape (skewness, distribution). Identify and explain any notable patterns or anomalies in the data.

**Main Question Q2 Trend Analysis**

Analyze how one or more key metrics change over time. Identify upward or downward trends, periods of growth or decline, and any seasonal or cyclical patterns. Use time-series line charts to visualize the trends.

**Main Question Q3 Cross-Tabulation Analysis**

Examine the interaction between two categorical dimensions simultaneously (e.g., product category x region, department x performance band). Use pd.crosstab() or pivot_table() and visualize the result as a heatmap to reveal patterns not visible in one-dimensional analysis.

**Main Question Q4 Drill-Down Analysis**

Start with a high-level aggregate view of a metric, then progressively decompose it into finer levels of detail (e.g., total sales -> by region -> by city -> by store). Use hierarchical groupby operations and interactive visualizations (e.g., Plotly) to enable the drill-down.

**Main Question Q5** Correlation & Predictive Analysis

Identify relationships between variables using a correlation matrix and heatmap. Select the strongest predictor(s) and build a simple predictive model (e.g., linear regression) to forecast or explain a target variable. Interpret the model output -- focus on insight over accuracy.

## Ad Hoc Questions

Ad hoc questions are follow-up questions that arise from your main analysis findings. They are reactive and investigative -- you discover them after seeing the data. Each ad hoc question must be derived from a different main question.

**Ad Hoc Q6** Follow-Up from Any Main Finding

Pose and answer a follow-up question triggered by a pattern or anomaly discovered in one of Q1-Q5. Describe what you found that prompted the question and how the analysis deepens your understanding of the business.

**Ad Hoc Q7** Follow-Up from a Different Main Finding

Pose and answer a second follow-up question derived from a different main question than Q6. This question should explore a different angle, segment, or dimension from the first ad hoc finding.

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# Project Phases & Requirements

## 01 Phase 1: Business Problem & Dataset Selection

Select a dataset and clearly define the scope of your analysis. Your submission for this phase must include:

- A description of the business domain and context
- Your chosen dataset: source, format, number of records and attributes
- A data dictionary summarizing key fields and their types
- Your seven (7) business questions, labeled Q1-Q5 and Ad Hoc Q6-Q7

## 02 Phase 2: Data Profiling & Exploratory Data Analysis (EDA)

Use Python (pandas, numpy) to explore and clean the data. Your notebook must include:

- Data loading and initial inspection (.shape, .dtypes, .info(), .describe())
- Missing value analysis and treatment (drop, impute, or flag)
- Outlier detection using IQR or Z-score method
- Distribution analysis for all numeric variables (histograms, boxplots)
- Correlation matrix for numeric variables
- Summary narrative: 3-5 key observations drawn from the EDA

## 03 Phase 3: Answering the Business Questions

For each of the seven business questions, provide the following in your Jupyter Notebook:

1. Python code that produces the analysis (clearly organized, with markdown cells explaining each step)

2. At least one visualization per question (choose the chart type appropriate to the question)
3. A written interpretation: what does the result mean for the business?

Required visualization types across the seven questions:

- Q1 -- Histogram or KDE plot
- Q2 -- Line chart (time series)
- Q3 -- Heatmap (crosstab)
- Q4 -- Hierarchical bar or treemap
- Q5 -- Scatter plot with regression line
- Q6 & Q7 -- Any chart appropriate to the finding

## 04 Phase 4: Dashboard

Create a consolidated dashboard that presents the most important findings from your analysis in a single view. Requirements:

- Minimum of four (4) charts arranged in a dashboard layout with filtering feature(s).
- Each chart must have a title and axis labels
- The dashboard must answer at least three of your main business questions
- Include a brief title and summary header that contextualizes the dashboard

## 05 Phase 5: Presentation

Deliver a presentation covering the following sections in order:

4. Business Overview -- Introduce the domain, dataset, and business questions
5. EDA Summary -- Key findings from data profiling; what surprised you
6. Business Questions -- Walk through each question with its chart and interpretation
7. Dashboard Walkthrough -- Demo or present the dashboard
8. Predictive Insight -- Explain the model, its output, and what it implies
9. Lessons Learned -- What you would do differently; what additional data you would want
10. Q&A

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# Deliverables & Submission

### Submission Format

Submit a single ZIP archive named Group_X_BusinessAnalytics.zip. The archive must contain all files listed below.

| File | Description | Format |
|---|---|---|
| Group_X_Notebook.ipynb | Jupyter notebook with all EDA, analysis, and business questions | .ipynb |
| Group_X_Dashboard.py / .html | Dashboard file (Plotly Dash script or exported HTML) | .py / .html |
| Group_X_Dataset.* | The dataset used (or a link to the source if too large) | .csv / .xlsx |

| | | |
|---|---|---|
| Group_X_Presentation.* | Slide deck for the final presentation | .pptx / .pdf |
| Group_X_DataDictionary.* | Data dictionary and business question definitions | .docx / .pdf |

## Recommended Tools & Libraries

| Purpose | Library | Key Functions / Notes |
|---|---|---|
| Data Wrangling | pandas, numpy | read_csv(), groupby(), crosstab(), pivot_table(), merge() |
| Statistical Analysis | scipy, statsmodels | describe(), pearsonr(), linregress(), OLS() |
| Visualization | matplotlib, seaborn | histplot(), heatmap(), lineplot(), boxplot(), regplot() |
| Interactive Charts | plotly | px.bar(), px.line(), px.scatter(), go.Heatmap() |
| Dashboard | plotly Dash or Panel | dcc.Graph(), html.Div(), layout, callbacks |
| Predictive | scikit-learn | LinearRegression(), train_test_split(), r2_score() |
| Notebook | Jupyter Notebook / Lab | Use markdown cells to narrate each analysis step |

## Tips for Success

- Choose a dataset with at least 50000 records and 15+ columns to ensure enough richness for all seven questions.
- Write markdown cells in your notebook that tell a story -- explain what you did and why, not just what the code does.
- Every chart must have a title, axis labels, and a one-sentence takeaway below it.
- The ad hoc questions should feel natural -- they should come from genuine curiosity about something you saw in the data.
- For the predictive component, focus on interpreting the result in business terms -- a clear insight with a simple model is better than a complex model with no explanation.
- Assign one group member as a Compliance Officer to verify all deliverable requirements are met before submission.